

Los 125 nodos sin interpretar

Generados desde la taxonomía de Mathlib y marcados **interpretado=False**: dicen *dónde vive* algo, no qué es categóricamente. Ordenados por peso — cuántos teoremas hay debajo — porque interpretar primero los grandes rinde más.

La columna que decide cuánto revisar es **«fuente»**. *docstring propio* y *Basic/Defs* son texto que un matemático escribió para ese concepto. *títulos del subárbol* describe qué hay dentro, no qué es. *sólo palabras* es el nombre y su vocabulario: éstos hay que mirarlos enteros.

Las **dependencias son reales** — salen del DAG de imports de Mathlib, que es acíclico y no lo escribió nadie a mano. Los **nombres están deducidos** de la ruta del módulo y NO comprobados con `#check`: por eso hoy no se inyectan al prompt.

fuelle de la descripción	cuántos	qué significa para la revisión
Basic	89	tomado de su fichero Basic · fiable
títulos del subárbol	24	sin docstring propio · describe el contenido, no el concepto
Defs	7	tomado de su fichero Defs · fiable
docstring propio	5	el módulo tiene su propio docstring · casi no necesita revisión

algebra — 60 nodos · 39476 teoremas

nodo · módulo de Mathlib	teoremas	fuelle	descripción	depende de	nombres que propone · sin verificar
algebra-order Mathlib.Algebra.Order	5140	títulos del subárbol	algebra order: Ordered scalar multiplication and vector addition; Ordered algebras; Chebyshev's sum inequality; Conditionally complete linear ordered fields; `Disjointed` for functions on a `SuccAddOrder`; Lemmas about `invOf` in	algebra-bigoperators, algebra-directsum, algebra-star, data-enat, data-ennreal, data-fin	Order, LinearOrderedAddCommMonoidWithTop, LinearOrderedAddCommGroupWithTop, IsOrderedVAdd, IsOrderedSMul
data-set Mathlib.Data.Set	2852	Basic	Basic properties of sets Sets in Lean are homogeneous; all their elements have the same type. Sets whose elements have type X are thus defined as $\text{Set } X := X \rightarrow \text{Prop}$. Note that this function need not be decidable. The definition is	algebra-bigoperators, algebra-order, data-enat, data-finset, data-list, data-nat	Set
algebra-polynomial Mathlib.Algebra.Polynomial	2380	Basic	Theory of univariate polynomials This file defines Polynomial R , the type of univariate polynomials over the semiring R , builds a semiring structure on it, and gives basic definitions that are expanded in other files in this dir	algebra-bigoperators, algebra-charp, algebra-monoidalgebra, algebra-mvpolynomial, algebra-order, data-enat	Polynomial, CAlgHom
data-finset Mathlib.Data.Finset	2292	Basic	Basic lemmas on finite sets This file contains lemmas on the interaction of various definitions on the Finset type. For an explanation of Finset design decisions, please see Mathlib/Data/Finset/Defs.lean . Main declarations Main d	algebra-bigoperators, algebra-order, data-fin, data-finsupp, data-int, data-list	Finset
data-nat Mathlib.Data.Nat	1916	Basic	Basic operations on the natural numbers This file builds on Mathlib/Data/Nat/Init.lean by adding basic lemmas on natural numbers depending on Mathlib definitions. See note [foundational algebra order theory].	algebra-bigoperators, algebra-charp, algebra-order, algebra-polynomial, data-enat, data-fin	Nat, AtLeastTwo
data-list Mathlib.Data.List	1835	Basic	Definitions on lists This file contains various definitions on lists. It does not contain proofs about these definitions, those are contained in other files in Data.List	algebra-bigoperators, algebra-order, data-fin, data-multiset, data-nat, data-set	List, AList, Disjoint, Cycle, Mem
algebra-lie Mathlib.Algebra.Lie	1228	Basic	Lie algebras This file defines Lie rings and Lie algebras over a commutative ring together with their modules, morphisms and equivalences, as well as various lemmas to make these definitions usable. Main definitions LieRing LieAlg	algebra-directsum, algebra-order, algebra-polynomial, data-finset, data-int, data-matrix	Lie, IsTrivial, IsLieAbelian, TrivialLieModule, LieRing
algebra-bigoperators Mathlib.Algebra.BigOperators	1100	títulos del subárbol	algebra big operators: Products of associated, prime, and irreducible elements.; Balancing a function; Average over a finset; Results about big operators with values in a field; Big operators and `Fin`; Finite products and sums ov	algebra-charp, algebra-order, data-finset, data-finsupp, data-int, data-list	BigOperators
algebra-category Mathlib.Algebra.Category	999	títulos del subárbol	algebra category: The category of Boolean rings; The category of commutative bialgebras over a commutative ring; The category of groups with zero; Category instance for algebras over a commutative ring; The category of R-algebras	algebra-bigoperators, algebra-directsum, algebra-mvpolynomial, categorytheory-adjunction, categorytheory-limits, categorytheory-monoidal	Category, BoolRing, Hom, CommBialgCat, GrpWithZero
ringtheory-polynomial Mathlib.RingTheory.Polynomial	884	Basic	Ring-theoretic supplement of Algebra.Polynomial. Main results MvPolynomial.isDomain : If a ring is an integral domain, then so is its polynomial ring over finitely many variables. Polynomial.isNoetherianRing : Hilbert basis theorem	algebra-charp, algebra-monoidalgebra, algebra-mvpolynomial, algebra-polynomial, analysis-complex, analysis-specialfunctions	Polynomial, IsPrimitive
data-multiset Mathlib.Data.Multiset	867	Basic	Multisets Multisets are finite sets with duplicates allowed. They are implemented here as the quotient of lists by permutation. This gives them computational content. This file contains the definition of Multiset and the basic pre	algebra-bigoperators, algebra-order, data-dfinsupp, data-list, data-nat, data-set	Multiset, Mem, Subset, Pairwise, Nodup

nodo · módulo de Mathlib	teoremas	fuelle	descripción	depende de	nombres que propone · sin verificar
algebra-mvpolynomial Mathlib.Algebra.MvPolynomial	755	Basic	Multivariate polynomials This file defines polynomial rings over a base ring (or even semiring), with variables from a general type σ (which could be infinite). Important definitions Let R be a commutative ring (or a semiring) and	algebra-bigoperators, algebra-monoidalgebra, algebra-order, algebra-polynomial, data-fin, data-finsupp	MvPolynomial, ACounit
algebra-star Mathlib.Algebra.Star	741	Basic	Star monoids, rings, and modules We introduce the basic algebraic notions of star monoids, star rings, and star modules. A star algebra is simply a star ring that is also a star module. These are implemented as "mixin" typeclasses	algebra-bigoperators, algebra-charp, algebra-order, data-int, data-rat, data-real	Star, StarMemClass, InvolutiveStar, TrivialStar, StarMul
grouptheory-perm Mathlib.GroupTheory.Perm	681	Basic	Extra lemmas about permutations This file proves miscellaneous lemmas about Equiv.Perm . TODO Most of the content of this file was moved to Mathlib/Algebra/Group/End.lean in https://github.com/leanprover-community/mathlib4/pull/22	algebra-order, data-finset, data-int, data-list, data-set, grouptheory-groupaction	Perm, Basis, Disjoint, IsSwap
data-matrix Mathlib.Data.Matrix	665	Basic	Matrices This file contains basic results on matrices including bundled versions of matrix operators. Implementation notes For convenience, Matrix $m \ n \ \alpha$ is defined as $m \rightarrow n \rightarrow \alpha$, as this allows elements of the matrix to be accesse	algebra-bigoperators, algebra-order, data-fin, data-finset, data-int, data-nat	Matrix, IsTwoBlockDiagonal, IsSimplyLaced
data-fin Mathlib.Data.Fin	660	Basic	The finite type with n elements Fin n is the type whose elements are natural numbers smaller than n . This file expands on the development in the core library. Main definitions finZeroElim : Elimination principle for the empty set	algebra-bigoperators, data-finset, data-int, data-nat, logic-equiv, logic-function	Fin, IsLT
data-ennreal Mathlib.Data.ENNReal	649	Basic	Extended non-negative reals We define ENNReal = $\mathbb{R}_{\geq 0}^{\infty} := \text{WithTop } \mathbb{R}_{\geq 0}$ to be the type of extended nonnegative real numbers, i.e., the interval $[0, +\infty]$. This type is used as the codomain of a MeasureTheory.Measure , and of the extend	algebra-bigoperators, algebra-order, data-finset, data-nnreal, order-interval	ENNReal, HolderTriple, HolderConjugate
data-finsupp Mathlib.Data.Finsupp	617	Basic	Miscellaneous definitions, lemmas, and constructions using finsupp Main declarations Finsupp.graph : the finset of input and output pairs with non-zero outputs. Finsupp.mapRange.equiv : Finsupp.mapRange as an equiv. Finsupp.mapDom	algebra-bigoperators, algebra-order, data-dfinsupp, data-finset, data-list, data-multiset	Finsupp, Lex, MonomialOrder
linearalgebra-rootsystem Mathlib.LinearAlgebra.RootSystem	590	Basic	Root data and root systems This file contains basic results for root systems and root data. Main definitions / results: RootPairing.ext : In characteristic zero if there is no torsion, the correspondence between roots and coroots	algebra-lie, linearalgebra-dimension, linearalgebra-span, order-interval	RootSystem, Base, IsPos, IsBalanced, RootPairing
grouptheory-groupaction Mathlib.GroupTheory.GroupAction	566	Basic	Basic properties of group actions This file primarily concerns itself with orbits, stabilizers, and other objects defined in terms of actions. Despite this file being called basic , low-level helper lemmas for algebraic manipulat	algebra-bigoperators, algebra-order, data-fin, data-nat, data-set, grouptheory-specificgroups	GroupAction, IsFixedBlock, IsInvariantBlock, IsTrivialBlock, IsBlock
ringtheory-powerseries Mathlib.RingTheory.PowerSeries	554	Basic	Formal power series (in one variable) This file defines (univariate) formal power series and develops the basic properties of these objects. A formal power series is to a polynomial like an infinite sum is to a finite sum. Formal	algebra-charp, algebra-mvpolynomial, algebra-polynomial, analysis-rclike, data-enat, data-finsupp	PowerSeries, HasEval, IsRestricted
ringtheory-localization Mathlib.RingTheory.Localization	551	Basic	Localizations of commutative rings This file contains various basic results on localizations. We characterize the localization of a commutative ring R at a submonoid M up to isomorphism; that is, a commutative ring S is the local	algebra-bigoperators, algebra-order, linearalgebra-basis, linearalgebra-dimension, ringtheory-spectrum	Localization, IsLocalization', IsLocalization, IsFractionRing, FractionRing
linearalgebra-tensorproduct Mathlib.LinearAlgebra.TensorProduct	546	Basic	Universal property of the tensor product Given any bilinear map $f : M \rightarrow \mathbb{R}[\sigma] N \rightarrow \mathbb{R}[\sigma] P$, there is a unique semilinear map TensorProduct.lift $f : \text{TensorProduct } R \ M \ N \rightarrow \mathbb{R}[\sigma] P$ whose composition with the canonical bilinear m	algebra-directsum, data-int, linearalgebra-basis, linearalgebra-finsupp, ringtheory-gradedalgebra, ringtheory-tensorproduct	TensorProduct, CompatibleSMul

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data-int MathLib.Data.Int	510	Basic	Basic operations on the integers This file builds on Data.Int.Init by adding basic lemmas on integers. depending on Mathlib definitions.	algebra-order, data-nat, data-nnreal, data-rat, data-set, logic-function	Int, ModEq
ringtheory-mvpolynomial MathLib.RingTheory.MvPolynomial	404	docstring propio	Multivariate polynomials over fields This file contains basic facts about multivariate polynomials over fields, for example that the dimension of the space of multivariate polynomials over a field is equal to the cardinality of fi	algebra-bigoperators, algebra-charp, algebra-directsum, algebra-monoidalgebra, algebra-mvpolynomial, algebra-order	MvPolynomial, IsHomogeneous, Monic, IsWeightedHomogeneous, NonTorsionWeight
ringtheory-mvpowerseries MathLib.RingTheory.MvPowerSeries	403	Basic	Formal (multivariate) power series This file defines multivariate formal power series and develops the basic properties of these objects. A formal power series is to a polynomial like an infinite sum is to a finite sum. We provide	algebra-charp, algebra-mvpolynomial, algebra-order, data-enat, data-finsupp, order-filter	MvPowerSeries, HasEval, IsWeightedHomogeneous, IsHomogeneous, HasSubst
data-real MathLib.Data.Real	399	Basic	Real numbers from Cauchy sequences This file defines $\mathbb{R}$ as the type of equivalence classes of Cauchy sequences of rational numbers. This choice is motivated by how easy it is to prove that $\mathbb{R}$ is a commutative ring, by simply lifting	algebra-order, algebra-star, data-enat, data-ennreal, data-nnreal, data-rat	Real, IsPowMul, IsBoundedWrt, HolderTriple, HolderConjugate
ringtheory-hahnseries MathLib.RingTheory.HahnSeries	373	Basic	Hahn Series If Γ is ordered and R has zero, then the type HahnSeries Γ R , which we denote as $R[[\Gamma]]$, consists of formal series over Γ with coefficients in R , whose supports are partially well-ordered. With further structure on R	algebra-bigoperators, algebra-order, analysis-rclike, data-finset, data-finsupp, data-rat	HahnSeries, HahnModule, SummableFamily, Equiv
linearalgebra-dimension MathLib.LinearAlgebra.Dimension	372	Basic	Dimension of modules and vector spaces Main definitions The rank of a module is defined as Module.rank : Cardinal . This is defined as the supremum of the cardinalities of linearly independent subsets. Main statements LinearMap.ra	data-set, linearalgebra-basis, linearalgebra-tensorproduct, logic-equiv, ringtheory-localization	Dimension, HasRankNullity, IsQuadraticExtension
data-ereal MathLib.Data.EReal	370	Basic	The extended real numbers This file defines EReal, $\mathbb{R}_{\infty}$ with a top element $\top$ and a bottom element $\perp$, implemented as WithBot (WithTop $\top$). EReal is a CompleteLinearOrder, deduced by typeclass inference from the fact that WithBot (W	data-ennreal, data-nat	EReal
ringtheory-spectrum MathLib.RingTheory.Spectrum	351	títulos del subárbol	ring theory spectrum: Maximal spectrum of a commutative (semi)ring; Prime spectrum of a commutative (semi)ring; Chevalley's theorem; Chevalley's theorem with complexity bound; Constructible sets in the prime spectrum; Prime spectr	algebra-order, algebra-polynomial, data-dfinsupp, order-heyting, order-succpred, ringtheory-localization	Spectrum, MaximalSpectrum, PiLocalization, DegreeType, InductionObj
representationtheory-homological MathLib.RepresentationTheory.Homological	344	títulos del subárbol	representation theory homological: The group cohomology of a k -linear G -representation; Group cohomology of a finite cyclic group; Functoriality of group cohomology; Hilbert's Theorem 90; Long exact sequence in group cohomolog	algebra-category, ringtheory-integralclosure	Homological, H1InfRes, IsCocycle $\mathbb{Z}$, IsCocycle $\mathbb{Z}$, IsCoboundary $\mathbb{Z}$
ringtheory-dedekinddomain MathLib.RingTheory.DedekindDomain	344	Basic	Dedekind rings and domains This file defines the notion of a Dedekind ring (domain), as a Noetherian integrally closed commutative ring (domain) of Krull dimension at most one. Main definitions - IsDedekindRing defines a Dedekind	algebra-order, algebra-polynomial, data-int, order-filter, ringtheory-fractionalideal, ringtheory-polynomial	DedekindDomain, DimensionLEOne, IsDedekindRing, IsDedekindDomain, IsDedekindDomainDvr
data-dfinsupp MathLib.Data.DFinsupp	331	Defs	Dependent functions with finite support For a non-dependent version see Mathlib/Data/Finsupp/Defs.lean . Notation This file introduces the notation $\prod_{i \in \alpha} a_i$ as notation for $DFinsupp \beta$, mirroring the $\alpha \rightarrow \beta$ notation used for Fins	algebra-bigoperators, algebra-order, data-finsupp, data-set, order-conditionallycomplete, order-interval	DFinsupp, Lex
data-seq MathLib.Data.Seq	330	Basic	Basic properties of sequences (possibly infinite lists) This file provides some basic lemmas about possibly infinite lists represented by the type Stream'.Seq .	data-enat, data-nat	Seq, Computation, Bisim0, IsBisimulation, Mem
data-rat MathLib.Data.Rat	308	Defs	Basics for the Rational Numbers Summary We define the integral domain structure on $\mathbb{Q}$ and prove basic lemmas about it. The definition of the field structure on $\mathbb{Q}$ will be done in Mathlib/Algebra/Field/Rat.lean once the Field class h	algebra-bigoperators, algebra-order, data-int, data-nat, data-pnat, data-real	Rat, NNRat, NNRatCast

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algebra-monoidalgebra MathLib.Algebra.MonoidAlgebra	303	Basic	Algebra structure on monoid algebras	algebra-directsum, data-finsupp, linearalgebra-basis, linearalgebra-finsupp, order-filter, ringtheory-gradedalgebra	MonoidAlgebra, Monic
ringtheory-gradedalgebra MathLib.RingTheory.GradedAlgebra	302	Basic	Internally-graded rings and algebras This file defines the typeclass GradedAlgebra $\mathbf{A}$, for working with an algebra A that is internally graded by a collection of submodules $\mathbf{A} : \iota \rightarrow \text{Submodule } R \mathbf{A}$. See the docstring of that typecla	algebra-directsum, linearalgebra-finsupp, linearalgebra-tensorproduct, order-filter, ringtheory-localization, ringtheory-tensorproduct	GradedAlgebra, GradedAlgHom, GradedRing, NumDenSameDeg, HomogeneousLocalization
ringtheory-wittvector MathLib.RingTheory.WittVector	299	Basic	Witt vectors This file verifies that the ring operations on WittVector $p \ R$ satisfy the axioms of a commutative ring. Main definitions WittVector.map : lifts a ring homomorphism $R \rightarrow S$ to a ring homomorphism $\mathbf{A} R \rightarrow \mathbf{A} S$. WittVector	algebra-charp, algebra-mypolynomial, algebra-order, data-nat	WittVector, IsPoly, IsPoly $\mathbf{A}$, Isocrystal, IsocrystalHom
linearalgebra-cliffordalgebra MathLib.LinearAlgebra.CliffordAlgebra	266	Basic	Clifford Algebras We construct the Clifford algebra of a module M over a commutative ring R, equipped with a quadratic form Q. Notation The Clifford algebra of the R-module M equipped with a quadratic form Q is an R-algebra de	algebra-category, algebra-star, grouptheory-groupaction, linearalgebra-dual, linearalgebra-tensorproduct, ringtheory-gradedalgebra	CliffordAlgebra, EvenHom
ringtheory-tensorproduct MathLib.RingTheory.TensorProduct	249	Basic	The tensor product of R-algebras This file provides results about the multiplicative structure on $A \otimes [R] B$ when R is a commutative (semi)ring and A and B are both R-algebras. On these tensor products, multiplication is characteri	algebra-category, algebra-monoidalgebra, algebra-mypolynomial, algebra-star, linearalgebra-basis, linearalgebra-finsupp	TensorProduct, IsEffective
algebra-quaternion MathLib.Algebra.Quaternion	241	docstring propio	Quaternions In this file we define quaternions $\mathbf{A}[R]$ over a commutative ring R, and define some algebraic structures on $\mathbf{A}[R]$. Main definitions QuaternionAlgebra R a b c, $\mathbf{A}[R, a, b, c]$: [Bourbaki, Algebra I][bourbaki1989] with	algebra-star, linearalgebra-dimension	Quaternion, QuaternionAlgebra
algebra-charp MathLib.Algebra.CharP	240	Basic	Characteristic of semirings This file collects some fundamental results on the characteristic of rings that don't need the extra imports of Mathlib/Algebra/CharP/Lemmas.lean. As such, we can probably reorganize and find a better	algebra-bigoperators, data-int, data-nat, grouptheory-perm, order-interval, ringtheory-localization	CharP, MixedCharZero
data-pnat MathLib.Data.PNat	239	Basic	The positive natural numbers This file develops the type $\mathbf{A}+$ or PNat, the subtype of natural numbers that are positive. It is defined in Data.PNat.Defs, but most of the development is deferred to here so that Data.PNat.Defs can h	algebra-bigoperators, algebra-order, data-int, data-multiset, data-nat, logic-equiv	PNat, PrimeMultiset, Prime, Coprime, XgcdType
ringtheory-fractionalideal MathLib.RingTheory.FractionalIdeal	238	Basic	Fractional ideals This file defines fractional ideals of an integral domain and proves basic facts about them. Main definitions Let S be a submonoid of an integral domain R and P the localization of R at S. IsFractional defines w	ringtheory-dedekinddomain, ringtheory-integralclosure, ringtheory-localization	FractionalIdeal, IsFractional
grouptheory-specificgroups MathLib.GroupTheory.SpecificGroups	226	títulos del subárbol	group theory specific groups: Alternating Groups; Cyclic groups; Dihedral Groups; Klein Four Group; Quaternion Groups; Centralizer of an element in the alternating group	algebra-order, data-nat, grouptheory-groupaction, grouptheory-perm, logic-equiv	SpecificGroups, IsAddKleinFour, IsKleinFour, IsZGroup
ringtheory-extension MathLib.RingTheory.Extension	225	Basic	Extension of algebras Main definitions - Algebra.Extension : An extension of an R-algebra S is an R-algebra P together with a surjection $P \rightarrow \mathbf{A}[R] R$. - Algebra.Extension.Hom : Given a commuting square $R \twoheadrightarrow P \twoheadrightarrow S \mid \downarrow R' \twoheadrightarrow P'$	algebra-mypolynomial, linearalgebra-basis, linearalgebra-tensorproduct, ringtheory-localization, ringtheory-mypolynomial, ringtheory-tensorproduct	Extension, Hom, Cotangent, Generators, Ring

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data-sym Mathlib.Data.Sym	223	Basic	Symmetric powers This file defines symmetric powers of a type. The nth symmetric power consists of homogeneous n-tuples modulo permutations by the symmetric group. The special case of 2-tuples is called the symmetric square, which	algebra-bigoperators, algebra-order, data-finset, data-finsupp	Sym, Sym', Sym2, Mem, IsDiag
algebra-directsum Mathlib.Algebra.DirectSum	221	Basic	Direct sum This file defines the direct sum of abelian groups, indexed by a discrete type. Notation $\bigoplus_i \beta_i$ is the n-ary direct sum DirectSum . This notation is in the DirectSum locale, accessible after open DirectSum . Reference	algebra-order, data-dfinsupp, data-finsupp, linearalgebra-basis	DirectSum, GAlgebra, IsInternal, Decomposition, IsHomogeneous
fieldtheory-ratfunc Mathlib.FieldTheory.RatFunc	218	Basic	The field structure of rational functions Main definitions Working with rational functions as polynomials: - RatFunc.instField provides a field structure You can use IsFractionRing API to treat RatFunc as the field of fractions of	algebra-charp, algebra-polynomial, ringtheory-dedekinddomain, ringtheory-localization, ringtheory-polynomial	RatFunc
linearalgebra-linearindependent Mathlib.LinearAlgebra.LinearIndependent	217	Basic	Linear independence This file collects basic consequences of linear (in)dependence and includes specialized tests for specific families of vectors. Main statements We prove several specialized tests for linear independence of fami	algebra-bigoperators, algebra-order, data-fin, linearalgebra-dual, linearalgebra-finsupp, logic-equiv	LinearIndependent, LinearIndepOn, Maximal
data-enat Mathlib.Data.ENat	214	Basic	Definition and basic properties of extended natural numbers In this file we define ENat (notation: $\mathbb{N}^\infty$) to be WithTop $\mathbb{N}$ and prove some basic lemmas about this type. Implementation details There are two natural coercions from $\mathbb{N}$ to	algebra-order, data-nat	ENat, ENatMap
data-num Mathlib.Data.Num	213	Basic	Binary representation of integers using inductive types Note: Unlike in Coq, where this representation is preferred because of the reliance on kernel reduction, in Lean this representation is discouraged in favor of the "Peano" na	algebra-order, data-int, data-nat	Num, Prime
linearalgebra-span Mathlib.LinearAlgebra.Span	211	Basic	The span of a set of vectors, as a submodule Submodule.span s is defined to be the smallest submodule containing the set s . Notation We introduce the notation $R \langle v \rangle$ for the span of a singleton, Submodule.span $R \{v\}$. This is $\backslash\text{spa}$	combinatorics-matroid, data-nat, data-sym, linearalgebra-dimension, linearalgebra-finsupp	Span, IsPrincipal
ringtheory-integralclosure Mathlib.RingTheory.IntegralClosure	209	títulos del subárbol	ring theory integral closure: Going down for integrally closed domains; Integrally closed rings; Integral closure of a subring.; Integral algebras; Integrality over ideals; Almost integral elements	algebra-polynomial, ringtheory-dedekinddomain, ringtheory-localization, ringtheory-polynomial, ringtheory-tensorproduct	IntegralClosure, IsIntegrallyClosedIn, IsIntegrallyClosed, IsIntegral, IsAlmostIntegral
linearalgebra-basis Mathlib.LinearAlgebra.Basis	207	Basic	Basic results on bases The main goal of this file is to show the equivalence between bases and families of vectors that are linearly independent and whose span is the whole space. There are also various lemmas on bases on specific	data-fin, linearalgebra-dual, linearalgebra-finsupp, linearalgebra-linearindependent, linearalgebra-span	Basis
data-nnreal Mathlib.Data.NNReal	204	Basic	Basic results on nonnegative real numbers This file contains all results on NNReal that do not directly follow from its basic structure. As a consequence, it is a bit of a random collection of results, and is a good target for cle	algebra-bigoperators, algebra-order, data-nat, data-real, order-conditionallycompletelattice	NNReal
linearalgebra-finsupp Mathlib.LinearAlgebra.Finsupp	203	Defs	Properties of the module $\alpha \rightarrow \mathbb{M}$ Given an R -module M , the R -module structure on $\alpha \rightarrow \mathbb{M}$ is defined in Mathlib/Data/Finsupp/SMul.lean . In this file we define LinearMap versions of various maps: Finsupp.single a : $M \rightarrow \mathbb{M}[R] \mapsto \mathbb{M}$	algebra-bigoperators, algebra-monoidalgebra, algebra-order, algebra-polynomial, data-finsupp, linearalgebra-basis	Finsupp
linearalgebra-dual Mathlib.LinearAlgebra.Dual	201	Defs	Dual vector spaces The dual space of an R\$-module \$M\$ is the R\$-module of R\$-linear maps \$M \to R\$. Main definitions Duals and transposes: Module.Dual R M defines the dual space of the R -module M , as $M \rightarrow \mathbb{M}[R] \rightarrow R$. Module.dualPa	linearalgebra-basis, linearalgebra-dimension, linearalgebra-span, ringtheory-tensorproduct	Dual, DualBases, IsReflexive

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data-sum Mathlib.Data.Sum	200	Basic	Additional lemmas about sum types Most of the former contents of this file have been moved to Batteries.	data-finsset, logic-function, order-heyting, order-hom, order-interval	Sum

analysis — 11 nodos · 9959 teoremas

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analysis-specialfunctions Mathlib.Analysis.SpecialFunctions	3906	títulos del subárbol	analysis special functions: Inverse of the cosh function; The arithmetic-geometric mean; Inverse of the sinh function; Inverse of the tanh function; Bernstein approximations and Weierstrass' theorem; Properties of Shannon q-ary en	algebra-bigoperators, algebra-order, algebra-polynomial, analysis-asymptotics, analysis-complex, analysis-convex	SpecialFunctions, IsExpCmpFilter
analysis-convex Mathlib.Analysis.Convex	1708	Basic	Convex sets In a $\mathbb{K}$ -vector space, we define the following property: Convex S : A set S is convex if for any two points $x, y \in S$ it includes segment $[x, y]$. We provide various equivalent versions, and prove that some specific sets	algebra-bigoperators, algebra-charp, algebra-order, analysis-locallyconvex, analysis-rclike, analysis-specialfunctions	Convex, AmpleSet, Wbtw, Sbtw, ConvexBody
analysis-complex Mathlib.Analysis.Complex	1366	Basic	Normed space structure on $\mathbb{C}$. This file gathers basic facts of analytic nature on the complex numbers. Main results This file registers $\mathbb{C}$ as a normed field, expresses basic properties of the norm, and gives tools on the real vecto	algebra-charp, algebra-order, algebra-polynomial, analysis-asymptotics, analysis-convex, analysis-locallyconvex	Complex, Circle
analysis-asymptotics Mathlib.Analysis.Asymptotics	761	Defs	Asymptotics We introduce these relations: $IsBigOWith c$ $f \sim g$: " f is big O of g along I with constant c "; $f = O[I]$ g : " f is big O of g along I "; $f = \Theta[I]$ g : " f is big O of g along I and vice versa"; $f = o[I]$ g : " f is little o of g "	algebra-polynomial, analysis-convex, analysis-locallyconvex, analysis-specialfunctions, topology-instances, topology-openpartialhomeomorph	Asymptotics, IsTheta, IsEquivalent
analysis-boxintegral Mathlib.Analysis.BoxIntegral	422	Basic	Integrals of Riemann, Henstock-Kurzweil, and McShane In this file we define the integral of a function over a box in $\mathbb{R}^n$. The same definition works for Riemann, Henstock-Kurzweil, and McShane integrals. As usual, we represent $\mathbb{R}^n$ as	algebra-bigoperators, data-nnreal, data-set, measuretheory-integral, order-interval, topology-order	BoxIntegral, HasIntegral, Integrable, Box, Icc
analysis-locallyconvex Mathlib.Analysis.LocallyConvex	365	Basic	Local convexity This file defines absorbent and balanced sets. An absorbent set is one that "surrounds" the origin. The idea is made precise by requiring that any point belongs to all large enough scalings of the set. This is the	analysis-convex, analysis-rclike, data-real, grouptheory-groupaction, linearalgebra-dual, linearalgebra-finsupp	LocallyConvex, AbsConvex, AbsConvexOpenSets, BarrelledSpace, Balanced
analysis-distribution Mathlib.Analysis.Distribution	314	títulos del subárbol	analysis distribution: Functions which vanish as distributions vanish as functions; Type classes for derivatives and the Laplacian; Distributions; Functions and measures of temperate growth; TemperedDistribution; Continuously diff	algebra-order, analysis-fourier, analysis-locallyconvex, analysis-specialfunctions, data-fin, data-nat	Distribution, ContDiffMapSupportedIn, ContDiffMapSupportedInClass, LineDeriv, LineDerivAdd
analysis-rclike Mathlib.Analysis.RCLike	311	Basic	$RCLike$: a typeclass for $\mathbb{R}$ or $\mathbb{C}$ This file defines the typeclass $RCLike$ intended to have only two instances: $\mathbb{R}$ and $\mathbb{C}$. It is meant for definitions and theorems which hold for both the real and the complex case, and in particular whe	algebra-bigoperators, algebra-order, analysis-specialfunctions, data-real, linearalgebra-basis, linearalgebra-dual	$RCLike$, $IsRCLikeNormedField$
analysis-real Mathlib.Analysis.Real	283	títulos del subárbol	analysis real: The cardinality of the reals; Representation of reals in positional system; Chudnovsky's formula for π ; 'Real.pi' is irrational; Leibniz's series for π ; The Wallis formula for π	algebra-order, analysis-complex, analysis-specialfunctions, data-rat, measuretheory-integral, order-filter	Real, Hyperreal, IsSt, Infinitesimal, InfinitePos

nodo · módulo de Mathlib	teoremas	fuentes	descripción	depende de	nombres que propone · sin verificar
analysis-fourier Mathlib.Analysis.Fourier	277	títulos del subárbol	analysis fourier: Fourier analysis on the additive circle; Multivariate Fourier series; Definition of BoundedContinuousFunction.char; The Fourier transform of the convolution; The Fourier transform; Derivatives of the Fourier tran	algebra-bigoperators, algebra-directsum, algebra-monoidalgebra, analysis-complex, analysis-distribution, analysis-rclike	Fourier, UnitAddTorus, FourierTransform, FourierTransformInv, FourierAdd
analysis-meromorphic Mathlib.Analysis.Meromorphic	246	Basic	Meromorphic functions Main statements: MeromorphicAt : definition of meromorphy at a point MeromorphicAt.iff_eventuallyEq_zpow_smul_analyticAt : f is meromorphic at $z \neq 0$ iff we have $f z = (z - z_0)^n \cdot g z$ on a punctured neighborho	algebra-order, analysis-specialfunctions	Meromorphic, MeromorphicAt, MeromorphicOn, Gamma, MeromorphicNFAt

set-theory — 13 nodos · 7777 teoremas

nodo · módulo de Mathlib	teoremas	fuentes	descripción	depende de	nombres que propone · sin verificar
order-filter Mathlib.Order.Filter	2192	Basic	Theory of filters on sets A filter on a type α is a collection of sets of α which contains the whole α , is upwards-closed, and is stable under intersection. They are mostly used to abstract two related kinds of ideas: limits, in	algebra-bigoperators, algebra-order, analysis-asymptotics, data-finset, data-int, data-nat	Filter, CardinalInterFilter, CountableInterFilter, HasCountableSeparatingOn, IsCountablyGenerated
order-interval Mathlib.Order.Interval	1916	Basic	Order intervals This file defines (nonempty) closed intervals in an order (see Set.Icc). This is a prototype for interval arithmetic. Main declarations NonemptyInterval : Nonempty intervals. Pairs where the second element is grea	algebra-order, data-finset, data-int, data-nat, data-set, order-booleanalgebra	Interval, NonemptyInterval, Icc, Ico, Ioc
order-hom Mathlib.Order.Hom	555	Basic	Order homomorphisms This file defines order homomorphisms, which are bundled monotone functions. A preorder homomorphism $f : \alpha \rightarrow \beta$ is a function $\alpha \rightarrow \beta$ along with a proof that $\forall x y, x \leq y \rightarrow f x \leq f y$. Main definitions In this fi	data-finset, data-set, data-sum, logic-equiv, logic-function, order-interval	Hom, OrderHom, OrderEmbedding, OrderIso, OrderHomClass
order-completelattice Mathlib.Order.CompleteLattice	468	Basic	Theory of complete lattices This file contains basic results on complete lattices. Naming conventions In lemma names, sSup is called sSup sInf is called sInf $\sup i$, $\inf i$ is called iSup $\inf i$, $\sup i$ is called iInf $\sup i$, $\inf i$ is called i	algebra-order, categorytheory-limits, data-finset, data-nat, data-set, order-bounds	CompleteLattice, CompleteSemilatticeSup, CompleteSemilatticeInf, CompleteLinearOrder
order-succpred Mathlib.Order.SuccPred	385	Basic	Successor and predecessor This file defines successor and predecessor orders. succ a, the successor of an element a : α is the least element greater than a. pred a is the greatest element less than a. Typical examples include $\mathbb{N}$	algebra-order, data-finset, data-nat, data-set, order-conditionallycompletelattice, order-interval	SuccPred, IsSuccArchimedean, IsPredArchimedean, NaiveSuccOrder, SuccOrder
order-upperlower Mathlib.Order.UpperLower	361	Basic	Properties of unbundled upper/lower sets This file proves results on IsUpperSet and IsLowerSet, including their interactions with set operations, images, preimages and order duals, and properties that reflect stronger assumptions	data-set, logic-equiv, order-hom, order-interval	UpperLower
settheory-zfc Mathlib.SetTheory.ZFC	329	Basic	A model of ZFC In this file, we model Zermelo-Fraenkel set theory (+ choice) using Lean's underlying type theory, building on the pre-sets defined in Mathlib/SetTheory/ZFC/PSet.lean. The theory of classes is developed in Mathlib/	data-fin, data-set	ZFC, ZFSet, Definable, Definable $\mathbb{N}$, Definable $\mathbb{R}$
order-monotone Mathlib.Order.Monotone	312	Basic	Monotonicity This file defines (strictly) monotone/antitone functions. Contrary to standard mathematical usage, "monotone"/"mono" here means "increasing", not "increasing or decreasing". We use "antitone"/"anti" to mean "decreasin	algebra-order, data-int, data-nat, data-set, logic-function, order-bounds	Monotone, Antitone, MonotoneOn, AntitoneOn, StrictMono
order-heyting Mathlib.Order.Heyting	280	Basic	Heyting algebras This file defines Heyting, co-Heyting and bi-Heyting algebras. A Heyting algebra is a bounded distributive lattice with an implication operation $\multimap$ such that $a \leq b \multimap c \Leftrightarrow a \wedge b \leq c$. It also comes with a pseudo-comp	order-booleanalgebra, order-hom	Heyting, GeneralizedHeytingAlgebra, GeneralizedCoheytingAlgebra, HeytingAlgebra, CoheytingAlgebra

nodo · módulo de Mathlib	teoremas	fuentes	descripción	depende de	nombres que propone · sin verificar
order-category Mathlib.Order.Category	268	títulos del subárbol	order category: The category of bounded distributive lattices; The category of bounded lattices; The category of bounded orders; The category of Boolean algebras; The category of complete lattices; The category of distributive lat	categorytheory-adjunction, categorytheory-limits, data-set, order-heyting, order-hom	Category, BddDistLat, Hom, BddLat, BddOrd
order-booleanalgebra MathLib.Order.BooleanAlgebra	252	Basic	Basic properties of Boolean algebras This file provides some basic definitions, functions as well as lemmas for functions and type classes related to Boolean algebras as defined in Mathlib/Order/BooleanAlgebra/Defs.lean . Referenc	data-set, order-heyting	BooleanAlgebra, GeneralizedBooleanAlgebra
order-conditionallycompletelattice MathLib.Order.ConditionallyCompleteLattice	248	Basic	Theory of conditionally complete lattices A conditionally complete lattice is a lattice in which every non-empty bounded subset s has a least upper bound and a greatest lower bound, denoted below by sSup s and sInf s . Typical exa	algebra-order, data-finsset, data-set, order-bounds	ConditionallyCompleteLattice, ConditionallyCompleteLinearOrder, ConditionallyCompleteLinearOrderBot
order-bounds MathLib.Order.Bounds	211	Basic	Upper / lower bounds In this file we prove various lemmas about upper/lower bounds of a set: monotonicity, behaviour under $\cup$, $\cap$, insert , and provide formulas for $\emptyset$, univ , and intervals.	data-set, order-hom, order-interval	Bounds, BddAbove, IsLeast, IsLUB, IsCofinalFor

category-theory — 9 nodos · 6158 teoremas

nodo · módulo de Mathlib	teoremas	fuentes	descripción	depende de	nombres que propone · sin verificar
categorytheory-limits MathLib.CategoryTheory.Limits	2834	títulos del subárbol	category theory limits: Limits and colimits in comma categories; Limits and the category of (co)cones; Cones and cocones; Connected limits; Creating (co)limits; Limits in the category of elements	algebra-category, categorytheory-adjunction, categorytheory-monoidal, categorytheory-morphismproperty, categorytheory-objectproperty, data-fin	Limits, Cone, Cocone, ConeMorphism, CoconeMorphism
categorytheory-monoidal MathLib.CategoryTheory.Monoidal	1316	títulos del subárbol	category theory monoidal: The category of bimodule objects over a pair of monoid objects.; The category of bimonoids in a braided monoidal category.; Monoidal categories; Half braidings and the Drinfeld center of a monoidal catego	algebra-category, categorytheory-adjunction, categorytheory-limits, categorytheory-objectproperty	Monoidal, Bimod, Hom, BimonObj, IsBimonHom
categorytheory-triangulated MathLib.CategoryTheory.Triangulated	400	Basic	Triangles This file contains the definition of triangles in an additive category with an additive shift. It also defines morphisms between these triangles. TODO: generalise this to n-angles in n-angulated categories as in https://	categorytheory-adjunction, categorytheory-limits, categorytheory-localization, categorytheory-morphismproperty, categorytheory-objectproperty, categorytheory-shift	Triangulated, IsTriangulated, Triangle, TriangleMorphism, IsHomological
categorytheory-localization MathLib.CategoryTheory.Localization	357	títulos del subárbol	category theory localization: Localization of adjunctions; Lifting of bifunctors; Bousfield localization; ObjectProperty.isLocal is stable under transfinite compositions; Calculus of fractions; Composition of localization functors	categorytheory-adjunction, categorytheory-limits, categorytheory-monoidal, categorytheory-morphismproperty, categorytheory-objectproperty, categorytheory-shift	Localization, IsInvertedBy■, Lifting■, LeftFraction, RightFraction
categorytheory-morphismproperty MathLib.CategoryTheory.MorphismProperty	353	Basic	Properties of morphisms We provide the basic framework for talking about properties of morphisms. The following meta-property is defined RespectsLeft P Q : P respects the property Q on the left if P f → P (i ■ f) where i satisfies	categorytheory-limits, categorytheory-objectproperty, order-interval	MorphismProperty, RespectsRight, RespectsLeft, Respects, RespectsIso
categorytheory-shift MathLib.CategoryTheory.Shift	236	Basic	Shift A Shift on a category C indexed by a monoid A is nothing more than a monoidal functor from A to C ■ C . A typical example to keep in mind might be the category of complexes ■ → C_{n-1} → C_n → C_{n+1} → ■ . It has a shift in	categorytheory-adjunction, categorytheory-limits, categorytheory-localization, categorytheory-monoidal, categorytheory-triangulated	Shift, CompatibilityUnit, CompatibilityCounit, CommShift, HasShift

nodo · módulo de Mathlib	teoremas	fuelle	descripción	depende de	nombres que propone · sin verificar
categorytheory-adjunction MathLib.CategoryTheory.Adjunction	228	Basic	Adjunctions between functors $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$. F is the left adjoint and G is the right adjoint. We provide various useful constructors: <code>mkOfHomEquiv mk' : co</code>	categorytheory-limits, categorytheory-morphismproperty, categorytheory-subobject	Adjunction, SolutionSetCondition, IsLeftAdjoint, IsRightAdjoint, CoreHomEquivUnitCounit
categorytheory-subobject MathLib.CategoryTheory.Subobject	228	Basic	Subobjects We define <code>Subobject X</code> as the quotient (by isomorphisms) of <code>MonoOver X := {f : Over X // Mono f.hom}</code> . Here <code>MonoOver X</code> is a thin category (a pair of objects has at most one morphism between them), so we can think of it as	categorytheory-adjunction, categorytheory-limits, categorytheory-objectproperty, data-finset, data-set	Subobject, IsArtinianObject, Factors, MonoOver, IsNoetherianObject
categorytheory-objectproperty MathLib.CategoryTheory.ObjectProperty	206	Basic	Properties of objects in a category Given a category $\mathcal{C}$, we introduce an abbreviation <code>ObjectProperty C</code> for predicates $\mathcal{C} \rightarrow \text{Prop}$. TODO refactor the file <code>Limits.FullSubcategory</code> in order to rename <code>ClosedUnderLimitsOfShape</code> as <code>ObjectPr</code>	categorytheory-adjunction, categorytheory-limits, categorytheory-morphismproperty, categorytheory-shift, order-booleanalgebra, order-completelattice	ObjectProperty, IsClosedUnderIsomorphisms, ColimitOfShape, IsClosedUnderColimitsOfShape, ContainsZero

topology — 11 nodos · 5398 teoremas

nodo · módulo de Mathlib	teoremas	fuelle	descripción	depende de	nombres que propone · sin verificar
topology-order MathLib.Topology.Order	1486	docstring propio	Ordering on topologies and (co)induced topologies Topologies on a fixed type α are ordered, by reverse inclusion. That is, for topologies $\mathbf{t}$ and $\mathbf{t}'$ on α , we write $\mathbf{t} \leq \mathbf{t}'$ if every set open in $\mathbf{t}$ is also open in $\mathbf{t}'$. (One also cal	data-ereal, data-set, order-filter, order-hom, order-interval, order-succpred	Order, IndiscreteTopology, NontrivialTopology, DiscreteTopology, OrderTopology
topology-instances MathLib.Topology.Instances	554	títulos del subárbol	topology instances: Ternary Cantor Set; Some results about the topology of $\mathbb{N}$; Instances related to the discrete topology; Topology on extended natural numbers; Topology on the integers; Topology of irrational numbers	algebra-bigoperators, algebra-order, analysis-complex, analysis-real, data-enat, data-ennreal	Instances, AddCircle, UnitAddCircle
topology-continuousmap MathLib.Topology.ContinuousMap	541	Basic	Continuous bundled maps In this file we define the type <code>ContinuousMap</code> of continuous bundled maps. We use the <code>DFunLike</code> design, so each type of morphisms has a companion typeclass which is meant to be satisfied by itself and all str	algebra-order, algebra-star, analysis-complex, analysis-rclike, analysis-specialfunctions, data-real	ContinuousMap, SeparatesPoints, SeparatesPointsStrongly, CocompactMap, CocompactMapClass
algebraictopology-simplicialset MathLib.AlgebraicTopology.SimplicialSet	520	Basic	Simplicial sets A simplicial set is just a simplicial object in <code>Type</code> , i.e. a <code>Type</code> -valued presheaf on the simplex category. (One might be tempted to call these "simplicial types" when working in type-theoretic foundations, but th	categorytheory-limits, categorytheory-monoidal, categorytheory-morphismproperty, data-fin, logic-equiv	SimplicialSet, SSet, Truncated, Augmented, Edge
topology-constructions MathLib.Topology.Constructions	395	docstring propio	Constructions of new topological spaces from old ones This file constructs <code>pi</code> types, subtypes and quotients of topological spaces and sets up their basic theory, such as criteria for maps into or out of these constructions to be c	data-fin, data-finset, order-filter	Constructions, IsDiscrete, CofiniteTopology
topology-category MathLib.Topology.Category	382	títulos del subárbol	topology category: The category of bornologies; Compactly generated topological spaces; Compacta and Compact Hausdorff Spaces; Delta-generated topological spaces; Category of finite topological spaces; The category of locales	algebra-category, categorytheory-adjunction, categorytheory-limits, categorytheory-monoidal, data-finset, data-set	Category, Born, CompactlyGenerated, Compactum, DeltaGenerated
topology-compactness MathLib.Topology.Compactness	366	títulos del subárbol	topology compactness: Topological bases in compact sets and compact spaces; Compact sets and compact spaces; Compactly coherent spaces and the k -ification; Compactly generated topological spaces; Delta-generated topological spaces	analysis-locallyconvex, data-ereal, data-finset, data-set, order-filter, topology-category	Compactness, CompactlyCoherentSpace, UCompactlyGeneratedSpace, CompactlyGeneratedSpace, DeltaGeneratedSpace

nodo · módulo de Mathlib	teoremas	fuelle	descripción	depende de	nombres que propone · sin verificar
topology-emetricspace Mathlib.Topology.EMetricSpace	348	Basic	Extended metric spaces Further results about extended metric spaces.	algebra-order, analysis-specialfunctions, data-ennreal, data-nat, order-interval, topology-compactness	EMetricSpace, BoundedVariationOn, LocallyBoundedVariationOn, EDist, PseudoEMetricSpace
topology-semicontinuity Mathlib.Topology.Semicontinuity	311	Basic	Lower and Upper Semicontinuity This file develops key properties of upper and lower semicontinuous functions. Main definitions and results We have some equivalent definitions of lower- and upper-semicontinuity (under certain restr	topology-compactness	Semicontinuity, SemicontinuousWithinAt, SemicontinuousOn, SemicontinuousAt, Semicontinuous
topology-sets MathLib.Topology.Sets	284	títulos del subárbol	topology sets: Closed sets; Compact open covered sets; Compact sets; Open covers; Open sets; Clopen upper sets	order-hom, topology-compactness, topology-continuousmap	Sets, Closeds, Clopens, IrreducibleCloseds, IsCompactOpenCovered
topology-openpartialhomeomorph MathLib.Topology.OpenPartialHomeomorph	211	Basic	Partial homeomorphisms: basic theory Main definitions OpenPartialHomeomorph.refl : the identity open partial homeomorphism Topology.IsOpenEmbedding.toOpenPartialHomeomorph : construct a partial homeomorphism from an open embedding	logic-equiv, topology-sets	OpenPartialHomeomorph, IsImage, EqOnSource

probability — 7 nodos · 4003 teoremas

nodo · módulo de Mathlib	teoremas	fuelle	descripción	depende de	nombres que propone · sin verificar
measuretheory-integral MathLib.MeasureTheory.Integral	1871	títulos del subárbol	measure theory integral: Bounding of integrals by asymptotics; Integral average of a function; Integration of bounded continuous functions; Circle Averages; Integral over a circle in $\mathbb{R}^n$; Circle integral transform	algebra-bigoperators, algebra-order, analysis-boxintegral, analysis-specialfunctions, measuretheory-group, order-filter	Integral, CircleIntegrable, FinMeasAdditive, DominatedFinMeasAdditive, StronglyMeasurableAtFilter
probability-independence MathLib.Probability.Independence	500	Basic	Independence of sets of sets and measure spaces (σ -algebras) A family of sets of sets $\pi : \iota \rightarrow \text{Set} (\text{Set } \Omega)$ is independent with respect to a measure μ if for any finite set of indices $s = \{i_1, \dots, i_n\}$, for any sets $f_{i_1} \in \pi_{i_1}$	measuretheory-group, measuretheory-integral	Independence, IndepSets, Indep, IndepSet, IndepFun
probability-moments MathLib.Probability.Moments	395	Basic	Moments and moment-generating function Main definitions ProbabilityTheory.moment X p μ : p th moment of a real random variable X with respect to measure μ , $\mu[X^p]$ ProbabilityTheory.centralMoment X p μ : p th central moment of X w	analysis-complex, analysis-locallyconvex, analysis-specialfunctions, measuretheory-integral, probability-independence	Moments, HasSubgaussianMGF, HasCondSubgaussianMGF
measuretheory-group MathLib.MeasureTheory.Group	374	Defs	Definitions about invariant measures In this file we define typeclasses for measures invariant under (scalar) multiplication. - MeasureTheory.SMulInvariantMeasure M α μ says that the measure μ is invariant under scalar multiplicat	analysis-convex, grouptheory-groupaction, measuretheory-integral, order-filter, topology-continuousmap	Group, MeasurableAdd, MeasurableAdd $\mathbb{N}$, MeasurableMul, MeasurableMul $\mathbb{N}$
probability-distributions MathLib.Probability.Distributions	340	títulos del subárbol	probability distributions: Beta distributions over $\mathbb{R}$; Cauchy Distribution over $\mathbb{R}$; Exponential distributions over $\mathbb{R}$; Fernique's theorem for rotation-invariant measures; Gamma distributions over $\mathbb{R}$; Geometric distributions over $\mathbb{N}$	algebra-order, analysis-specialfunctions, measuretheory-integral, probability-independence, probability-moments	Distributions, IsSetBernoulli, IsUniform, IsGaussian
measuretheory-vectormeasure MathLib.MeasureTheory.VectorMeasure	264	Basic	Vector-valued measures This file defines vector-valued measures, which are σ -additive functions from a set to an additive monoid M such that it maps the empty set and non-measurable sets to zero. In the case that M = $\mathbb{R}$, we called	topology-emetricspace	VectorMeasure, SignedMeasure, AbsolutelyContinuous, MutuallySingular, ExistsOneDivLT
measuretheory-outermeasure MathLib.MeasureTheory.OuterMeasure	259	Basic	Outer Measures An outer measure is a function $\mu : \text{Set } \alpha \rightarrow \mathbb{R}_{\geq 0}$, from the powerset of a type to the extended nonnegative real numbers that satisfies the following conditions: 1. $\mu \emptyset = 0$; 2. μ is monotone; 3. μ is countably subadd	algebra-order, data-ennreal, data-fin, topology-order	OuterMeasure, IsCaratheodory, OuterMeasureClass

geometry — 4 nodos · 2893 teoremas

nodo · módulo de Mathlib	teoremas	fuentes	descripción	depende de	nombres que propone · sin verificar
geometry-euclidean Mathlib.Geometry.Euclidean	1281	Basic	Euclidean spaces This file makes some definitions and proves very basic geometrical results about real inner product spaces and Euclidean affine spaces. Results about real inner product spaces that involve the norm and inner produ	analysis-convex, analysis-specialfunctions, topology-instances	Euclidean, ExcenterExists, OrthocentricSystem
algebraicgeometry-ellipticcurve Mathlib.AlgebraicGeometry.EllipticCurve	1030	títulos del subárbol	algebraic geometry elliptic curve: Elliptic curves with same j-invariants are isomorphic; Models of elliptic curves with prescribed j-invariant; Some normal forms of elliptic curves; Reduction of Weierstrass curves over local fiel	algebra-charp, algebra-mvpolynomial, algebra-polynomial, data-fin, ringtheory-polynomial	EllipticCurve, IsCharNeTwoNF, IsShortNF, IsCharThreeJNeZeroNF, IsCharTwoJNeZeroNF
algebraicgeometry-morphisms Mathlib.AlgebraicGeometry.Morphisms	379	Basic	Properties of morphisms between Schemes We provide the basic framework for talking about properties of morphisms between Schemes. A MorphismProperty Scheme is a predicate on morphisms between schemes. For properties local at the t	algebra-category, categorytheory-limits, categorytheory-morphismproperty, data-list, ringtheory-spectrum	Morphisms, IsAffineHom, IsZariskiLocalAtTarget, IsZariskiLocalAtSource, AffineTargetMorphismProperty
geometry-ringspace Mathlib.Geometry.RingedSpace	203	Basic	Ringed spaces We introduce the category of ringed spaces, as an alias for SheafedSpace CommRingCat . The facts collected in this file are typically stated for locally ringed spaces, but never actually make use of the locality of s	algebra-category, algebra-order, categorytheory-adjunction, categorytheory-limits, topology-category	RingedSpace, LocallyRingedSpace, Hom, IsOpenImmersion, SheafedSpace

combinatorics — 3 nodos · 1775 teoremas

nodo · módulo de Mathlib	teoremas	fuentes	descripción	depende de	nombres que propone · sin verificar
combinatorics-matroid Mathlib.Combinatorics.Matroid	1180	Basic	Matroids A Matroid is a structure that combinatorially abstracts the notion of linear independence and dependence; matroids have connections with graph theory, discrete optimization, additive combinatorics and algebraic geometry.	data-enat, data-set, order-interval, order-upperlower	Matroid, ExchangeProperty, ExistsMaximalSubsetProperty, Finite, Nonempty
combinatorics-additive Mathlib.Combinatorics.Additive	366	títulos del subárbol	combinatorics additive: Approximate subgroups; The Cauchy-Davenport theorem; Convolution; Relation of covering by cosets; Dissociation and span; Doubling and difference constants	algebra-bigoperators, algebra-charp, algebra-order, analysis-complex, analysis-fourier, data-fin	Additive, IsApproximateAddSubgroup, IsApproximateSubgroup, DevosMulRel, CovBySMul
combinatorics-setfamily Mathlib.Combinatorics.SetFamily	229	títulos del subárbol	combinatorics set family: The Ahlswede-Zhang identity; The four functions theorem and corollaries; Harris-Kleitman inequality; Intersecting families; Kleitman's bound on the size of intersecting families; Kruskal-Katona theorem	algebra-bigoperators, algebra-order, data-finset, data-nat, data-rat, logic-function	SetFamily, Intersecting, IsIntersectingOf, UsefulCompression, Shatters

number-theory — 3 nodos · 1199 teoremas

nodo · módulo de Mathlib	teoremas	fuentes	descripción	depende de	nombres que propone · sin verificar
numbertheory-modularforms Mathlib.NumberTheory.ModularForms	514	Basic	Modular forms This file defines modular forms and proves some basic properties about them. Including constructing the graded ring of modular forms. We begin by defining modular forms and cusp forms as extension of SlashInvariantFo	algebra-directsum, analysis-asymptotics, analysis-complex, analysis-rclike, analysis-specialfunctions, numbertheory-lseries	ModularForms, HasDetPlusMinusOne, HasDetOne, IsArithmetic, ModularForm
numbertheory-lseries Mathlib.NumberTheory.LSeries	485	Basic	L-series Given a sequence f: $\mathbb{N} \rightarrow \mathbb{C}$, we define the corresponding L-series. Main Definitions LSeries.term f s n is the n th term of the L-series of the sequence f at s: $\mathbb{C}$. We define it to be zero when n = 0 . LSeries f is the L-s	algebra-bigoperators, analysis-asymptotics, analysis-complex, analysis-fourier, analysis-specialfunctions, data-ereal	LSeries, WeakFEPair, StrongFEPair, LSeriesSummable, LSeriesHasSum

nodo · módulo de Mathlib	teoremas	fuelle	descripción	depende de	nombres que propone · sin verificar
numbertheory-arithmeticfunction Mathlib.NumberTheory.ArithmeticFunction	200	Defs	Arithmetic Functions and Dirichlet Convolution This file defines arithmetic functions, which are functions from $\mathbb{N}$ to a specified type that map 0 to 0. In the literature, they are often instead defined as functions from $\mathbb{N}^+$. These	algebra-order, analysis-specialfunctions, data-nat	ArithmeticFunction, Carmichael, IsMultiplicative

logic — 3 nodos · 1174 teoremas

nodo · módulo de Mathlib	teoremas	fuelle	descripción	depende de	nombres que propone · sin verificar
logic-equiv Mathlib.Logic.Equiv	659	Basic	Equivalence between types In this file we continue the work on equivalences begun in Mathlib/Logic/Equiv/Defs.lean , defining a lot of equivalences between various types and operations on these equivalences. More definitions of th	data-fin, data-finset, data-int, data-multiset, data-nat, data-set	Equiv, Perm
logic-function Mathlib.Logic.Function	306	Basic	Miscellaneous function constructions and lemmas	data-fin, data-nat, data-set, logic-basic	Function, IsPartialInv, FactorsThrough, HasUncurry, Involution
logic-basic Mathlib.Logic.Basic	209	docstring propio	Basic logic properties This file is one of the earliest imports in mathlib. Implementation notes Theorems that require decidability hypotheses are in the namespace Decidable . Classical versions are in the namespace Classical .	–	Basic, Fact, Xor'

computation — 1 nodos · 221 teoremas

nodo · módulo de Mathlib	teoremas	fuelle	descripción	depende de	nombres que propone · sin verificar
computability-primrec Mathlib.Computability.Primrec	221	Basic	The primitive recursive functions The primitive recursive functions are the least collection of functions $\mathbb{N} \rightarrow \mathbb{N}$ which are closed under projections (using the pair pairing function), composition, zero, successor, and primitive recu	algebra-order, logic-function	Primrec, Primcodable, Primrec $\mathbb{N}$, PrimrecPred, PrimrecRel